



Management Performance:

Hea 18 Legionella risk management



Credits

No Minimum Standard

Aim

To ensure that water systems are managed to minimise the risk from legionella.

Question

Has an assessment of the asset's water systems been performed to identify the potential risk of exposure to legionella bacteria and, if so, are management processes in place to prevent or control the risk?

Credits	Answer	Select a single answer option
0	A.	Question not answered
0	B.	No
4	C.	Yes

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	Filtering Where water systems are not present in the asset and no water from nearby facilities (e.g. adjacent building) will be used by the asset users, this issue can be filtered out. Also see Specific Note below.	All
2.	A competent person has performed a risk assessment that identifies and assesses the risk of exposure of asset users to legionella bacteria from the asset's water systems and any required prevention or control measures. All water systems that present a potential risk from legionella bacteria must be assessed, including, but not limited to: a) Hot and cold water storage and distribution systems b) Evaporative cooling systems, e.g. cooling towers and evaporative condensers c) Spa pools, hot tubs and whirlpool baths d) Fountains and water features e) Humidifiers	C

Criterion	Assessment criteria	Applicable Answer
	f) Emergency showers, eyebaths and face-wash fountains g) Industrial water systems, e.g. air washers, wet scrubbers, vehicle washers, wastewater treatment plants, misting devices and sprinkler systems. A water system includes all equipment and components associated with that system, e.g. all associated pipework, pumps, feed tanks, valves, showers, heat exchangers, quench tanks, water softeners, chillers, etc.	
3.	Where the risk assessment identifies that prevention or control measures are necessary, the following must be in place: a) A competent person or persons appointed to manage the identified risks and control measures (this could be an employee or a contractor). b) Documentation specifying the measures implemented to prevent or control the risk (see Methodology). c) Process and procedure(s) for the regular monitoring and inspection of the condition and performance of the water systems and control measures, and the remedial actions to be taken in the event of a failure in control measures. d) Written records including details of: • The findings of the risk assessment. • The prevention and control measures and their implementation. • The state of operation of each system, i.e. in use or not in use. • The results and dates of all monitoring, inspection, testing or checks of the system(s). • The competent person(s) and all personnel concerned with the operation and maintenance of the system(s) and their training records. Where a risk assessment shows that risks are low and that no prevention and control measures are required, it is not necessary to demonstrate compliance with this criterion (subject to compliance with criterion 4).	C
4.	The risk assessment must be reviewed periodically, especially following any changes to the water system(s) or changes to the asset's operation that may result in significant changes to water use within the asset, e.g. change in number of users or change in use of space(s).	C

Specific notes

Asset type specific	
1.	No facilities Where the asset under assessment does not contain any water systems, and asset users have access to water facilities (e.g. taps, toilets, showers, etc.) in an adjacent asset, the adjacent asset's water systems must demonstrate compliance with the assessment criteria.

Methodology

Legionella prevention and control measures

Prevention and control measures include, but are not limited to:

- a) Avoiding water temperatures between 20°C and 45°C and conditions that favour the growth of legionella bacteria and other microorganisms.
- b) Avoiding water stagnation which may encourage the growth of biofilm.
- c) Avoiding the use of materials that harbour bacteria and other microorganisms or provide nutrients for microbial growth.
- d) Controlling the release of water spray.
- e) Maintaining the cleanliness of the system and water in it.
- f) Using water treatment techniques, e.g. biocides, chlorination, heat.
- g) Ensuring the correct and safe operation and maintenance of the water system.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
2	Documentation in accordance with Assessment criterion 2.

Definitions

Legionella:

Genus of bacteria that can cause infection (Legionellosis) through the inhalation of contaminated water droplets or aerosols, which can range in severity from a mild, febrile illness (Pontiac fever) to a rapid and potentially fatal pneumonia (Legionnaires' disease).

Competent person:

A person that has sufficient authority, competence and knowledge of the asset to ensure that all operational procedures are carried out in a timely and effective manner, and has such ability, experience, instruction, information, training and resources to enable them to carry out their tasks competently and safely. In particular, they should know the:

- a) Potential sources of legionella bacteria and the risks they present;

- b) Measures to adopt, including the precautions to take to protect the people concerned, and their significance;
- c) Measures to take to ensure that the control measures remain effective, and their significance.

Additional information

Legionella and water systems guidance

Guidance on the risk assessment and control of legionella bacteria in water systems include:

- American Society of Heating, Refrigerating and Air-Conditioning Engineers; ANSI/ASHRAE Standard 188-2018 Legionellosis: Risk Management for Building Water Systems; 2018.
- American Society of Heating, Refrigerating and Air-Conditioning Engineers; ASHRAE Guideline 12-2000 - Minimizing the Risk of Legionellosis Associated with Building Water Systems; 2000.
- British Standards Institution; BS 8580-1:2019 Water quality – Risk assessments for Legionella control – Code of practice; 2019.
- The Chartered Institution of Building Services Engineers; CIBSE TM13: 2013 Minimising the risk of Legionnaires' disease; 2013.
- European Centre for Disease Prevention and Control; European Technical Guidelines for the Investigation, Control and Prevention of Travel Associated Legionnaires' Disease; 2017.
- World Health Organisation; Legionella and the Prevention of Legionellosis; 2007.